BR41N.IO

THE BRAIN-COMPUTER INTERFACE DESIGNERS HACKATHON

2026

NEUROTECH ASU

APRIL 25-26

PROJECTS SPECIFICATION

PROJECT (1)

EXPLORE THE UNIVERSE WTH YOUR BRAIN

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ASU

OVERVIEW

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Description: An immersive, hands-free space exploration platform that allows users to navigate a 3D universe using only their brain.

Category: Programming.

CORE MECHANISMS

- **Target Selection (SSVEP):** Star clusters and galaxies on the screen "twinkle" at specific, unique frequencies (e.g. 10Hz, 12Hz, 15Hz). When the user focuses their gaze on a specific star, the system detects the corresponding SSVEP in the visual cortex, triggering a "warp jump" or zoom-in to that location.

- **Directional Control (MI):** To rotate the camera, the user uses Motor Imagery (MI), imagining the movement of their left/right hand. The system classifies these internal motor intentions to provide smooth, continuous orientation in the 3D space.

PROJECT OBJECTIVES

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- Develop a beautiful & navigatable 3D Universe and deliver a fully functional prototype for the br41n.io hackathon.
- Successfully classify at least three distinct SSVEP frequencies (e.g., 10Hz, 12Hz, 15Hz) for "warp jump" navigation.
- Successfully classify left/right Motor Imagery to control camera rotation in a Unity environment.
- Establish a rapid calibration protocol to allow new users to use the system within minutes.

WINNING EDGE

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- Combining two BCI paradigms for seamless navigation (SSVEP & MI)
- Creative & Original application for SSVEP (flickering stars)
- Visual strikingness & beauty
- Translational potential & real-world impact → the navigation mechanisms can be easily adapted into any BCI navigation system, not just floating around the cosmos!

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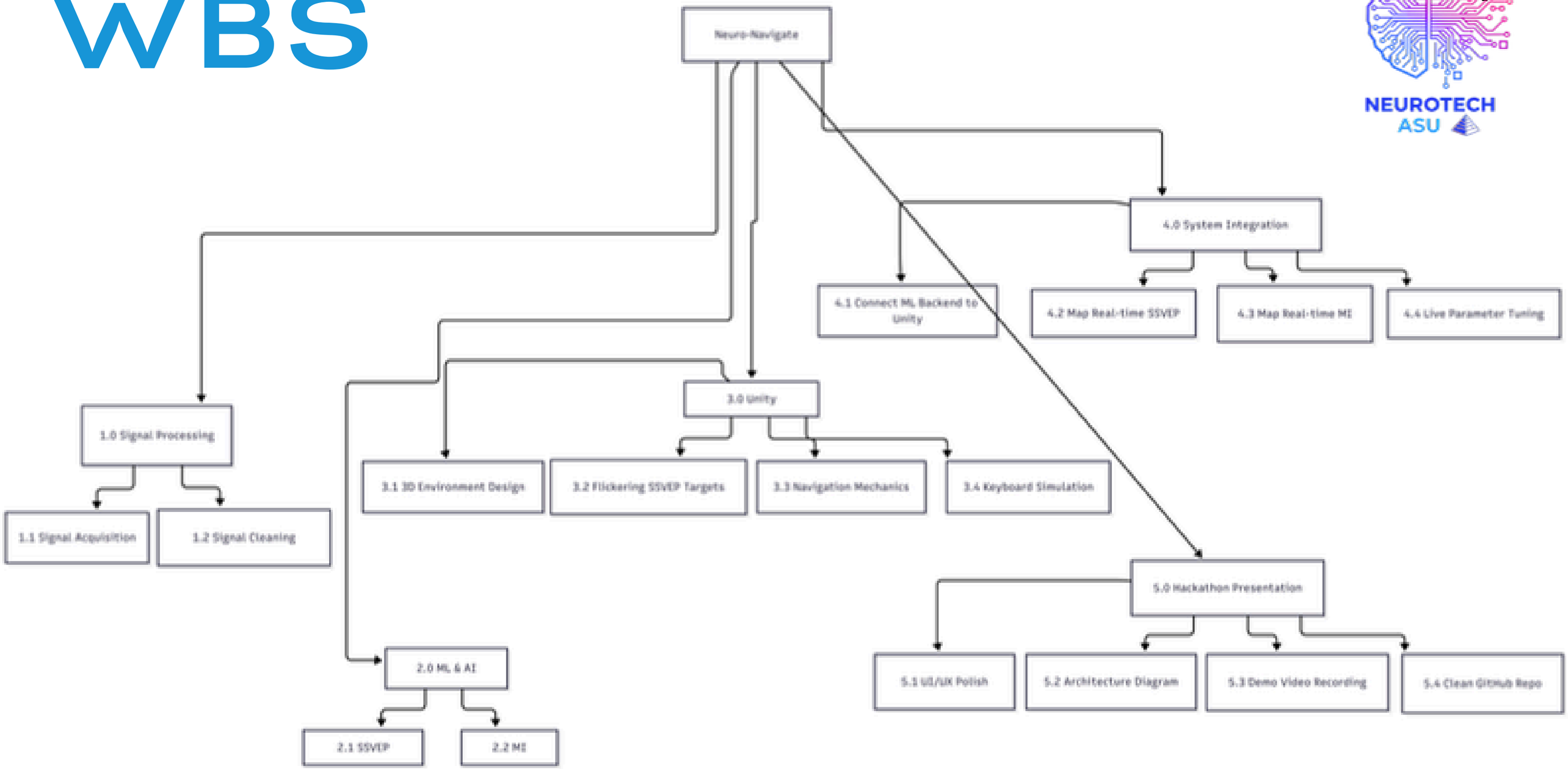
NEURO-NAVIGATE

- SIGNAL PROCESSING
- AI & ML
- UNITY & UX
- INTEGRATION & TESTING

WBS


Neuro-Navigate
1.0 Signal Processing
1.1 Signal Acquisition
1.2 Signal Cleaning
2.0 ML & AI
2.1 SSVEP
2.2 MI
3.0 Unity
3.1 3D Environment Design
3.2 Flickering SSVEP Targets
3.3 Navigation Mechanics
3.4 Keyboard Simulation
4.0 System Integration
4.1 Connect ML Backend to Unity
4.2 Map Real-time SSVEP
4.3 Map Real-time MI
4.4 Live Parameter Tuning
5.0 Hackathon Presentation
5.1 UI/UX Polish
5.2 Architecture Diagram
5.3 Demo Video Recording
5.4 Clean Github Repo

PROJECT (2): CONNECT YOUR BRAIN TO CHATGPT

OVERVIEW

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Description: A BCI communication system. It combines a neural speller with a mental state inference model, leveraging the ChatGPT API to predict user intent, speed up typing, and adapt AI responses based on the user's real-time cognitive state.

Category: Programming.

CORE MECHANISMS

- **LLM-Powered Predictive Spelling (EP):** The user spells only the first 1-2 letters of a word using an EP Grid. They are instantly sent to the ChatGPT API, which predicts the three most likely words based on the user's overarching task, displaying them as selectable targets (SSVEP).

- **Mental State Context (Bandpower Inference):** While the user spells, the system analyzes EEG rhythms to classify their cognitive load (e.g., "Highly Focused" vs. "Fatigued"). This inferred state is appended to the final prompt, instructing the LLM to dynamically adjust the tone and complexity of its response to match the user's mental state.

PROJECT OBJECTIVES

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- Develop a visually-pleasing Unity UI that integrates a standard BCI speller grid with a static row for three LLM-generated autocomplete suggestions.
- Successfully extract and classify EEG wave ratios to determine at least three distinct mental states (High Focus, Fatigue, Neutral) in real-time.
- Develop a backend Python API that merges the spelled text and the mental state context into a single prompt, and sends the response to the UI.
- Reduce standard BCI text-generation time through predictive autocomplete.

WINNING EDGE

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- A unique & creative approach for solving the famous problem of slow BCI spellers. (LLM + SSVEP powered autocomplete)
- Contextual Awareness: ChatGPT won't predict words randomly, but based on the topic at hand.
- Mental State Awareness: ChatGPT will tailor its response based on the user's mental state.

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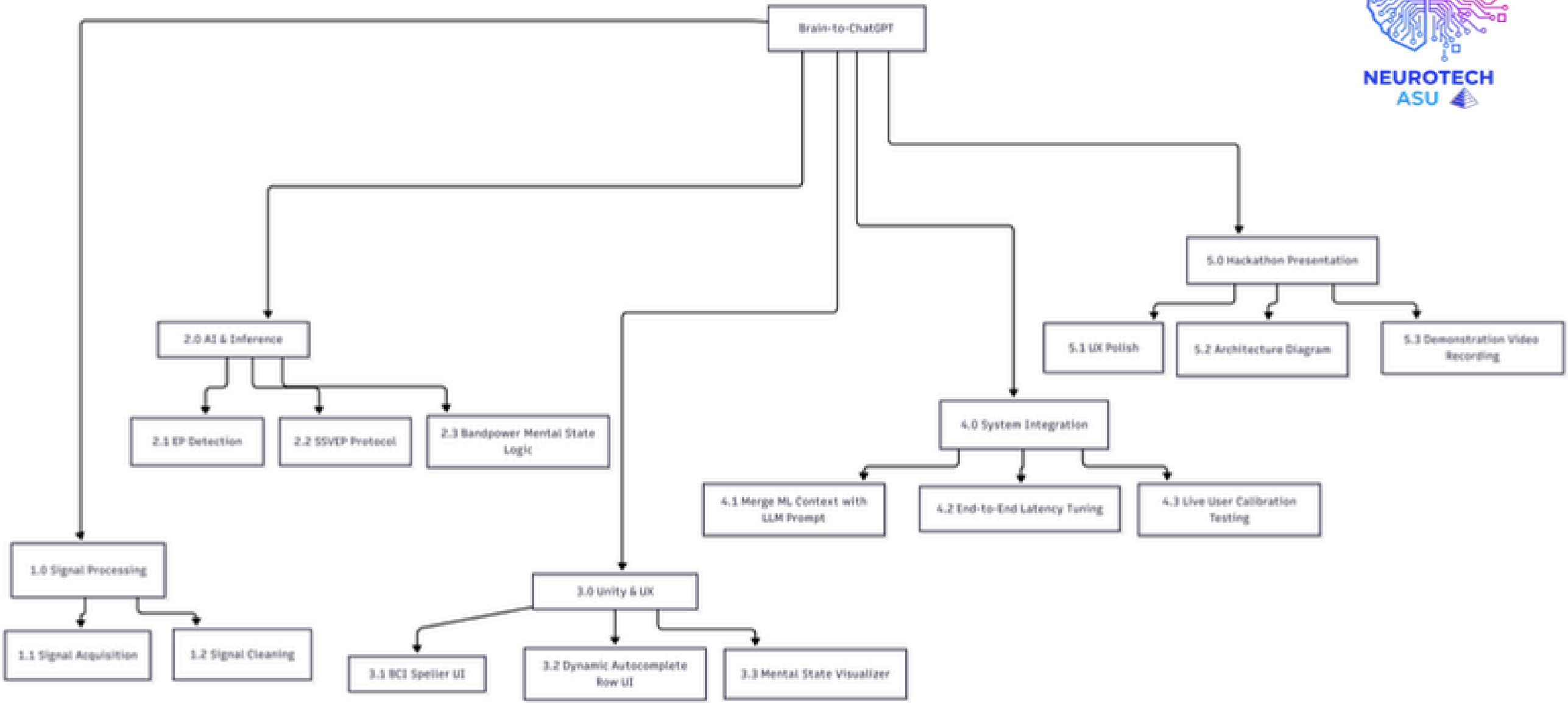
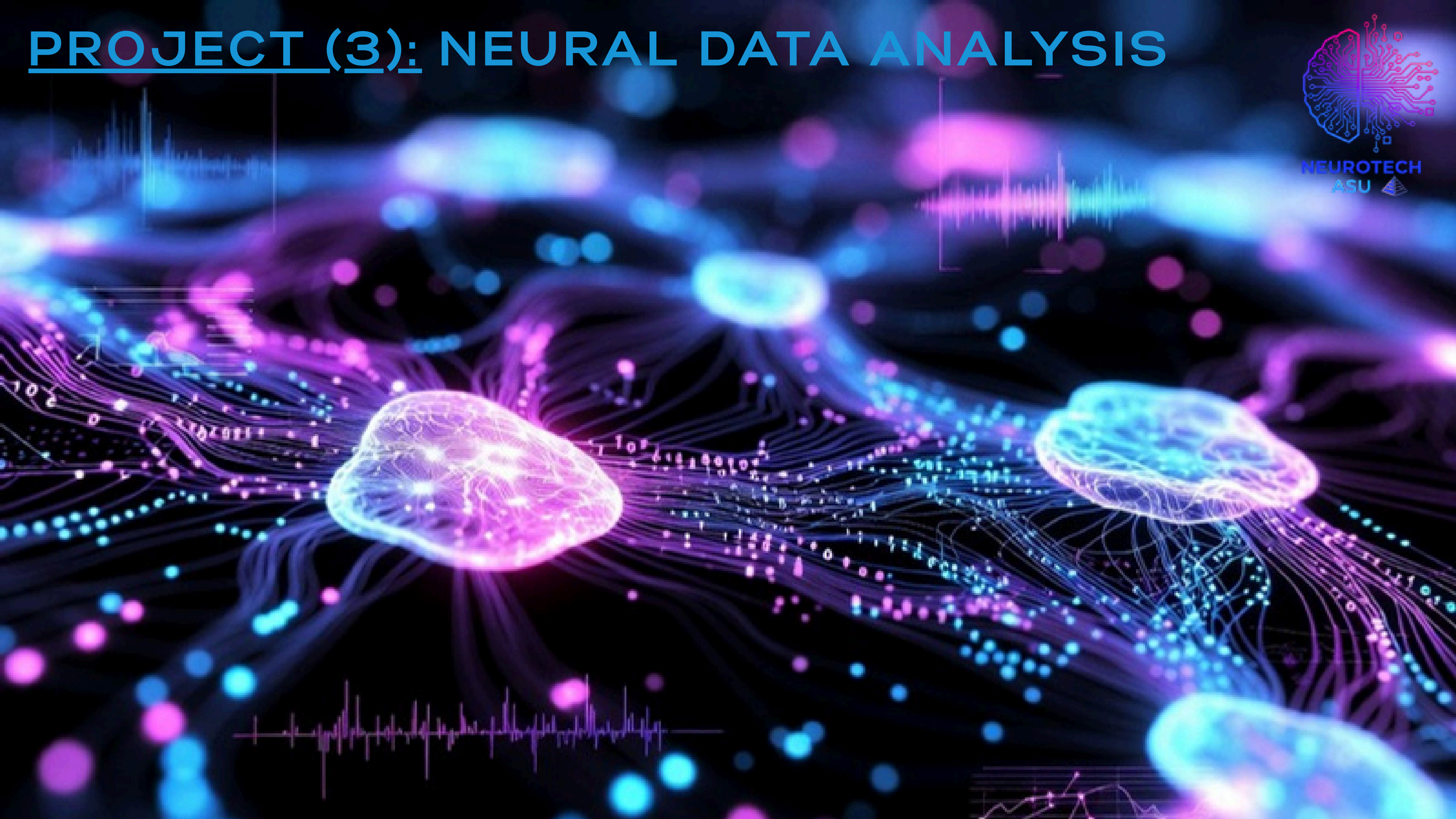
BRAIN<->CHATGPT

- SIGNAL PROCESSING
- BANDPOWER & EP INFERENCE
- UNITY & UX
- INTEGRATION & TESTING

WBS


Brain-to-ChatGPT
1.0 Signal Processing
1.1 Signal Acquisition
1.2 Signal Cleaning
2.0 AI & Inference
2.1 EP Detection
2.2 SSVEP Protocol
2.3 Bandpower Mental State Logic
3.0 Unity & UX
3.1 BCI Speller UI
3.2 Dynamic Autocomplete Row UI
3.3 Mental State Visualizer
4.0 System Integration
4.1 Merge ML Context with LLM Prompt
4.2 End-to-End Latency Tuning
4.3 Live User Calibration Testing
5.0 Hackathon Presentation
5.1 UX Polish
5.2 Architecture Diagram
5.3 Demonstration Video Recording

PROJECT (3): NEURAL DATA ANALYSIS



OVERVIEW

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Description: Analyze BCI dataset offered by g.tec during the hackathon. The goal is to optimize the signal processing pipeline: pre-processing, feature extraction, & classification, and compare the results with state-of-the-art techniques.

Category: Data Analysis

CORE MECHANISMS

- **Pre-processing Pipeline:** The system ingests the provided data and applies appropriate pre-processing techniques.

- **Feature Extraction:** Select the most expressive features for the dataset and the required classification task.

- **Classification:** Select a suitable ML model and optimize its parameters to achieve better than state-of-the-art results.

PROJECT OBJECTIVES

- Execute a complete data analysis pipeline in a purely software-based environment.
- Extract the most useful & expressive features.
- Achieve state-of-the-art classification accuracy.
- Document the performance of implemented algorithms against published research.

WINNING EDGE

- Extract useful & expressive features.
- Achieve better than state-of-the-art classification accuracy.
- Rigorous reporting & presentation.

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NEURO-ANALYSIS

SIGNAL PROCESSING

FEATURE EXTRACTION

ML/DL CLASSIFICATION

WBS

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GENERAL NOTES

- **Shared Foundation:** Several tasks overlap across projects (e.g., signal processing, SSVEP protocol). These foundational pipelines are our immediate priority.
- **The Hardware-Free Head Start:** We have one headset and many brilliant minds. Do not wait for hardware access. Teams can immediately begin building and testing pipelines using public datasets.
- **Communication is our API:** With parallel development, silence is a blocker. Make sure you “work out loud”, send your progress in the WP group, and co-work with your teammates on Discord!
- **The Hackathon Mindset:** We are building prototypes, not medical devices. Prioritize a low-latency proof-of-concept and a good demo over a perfect system. Fail fast, adjust, and keep moving.

LET'S WIN THIS!!

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g·tec